

Valkyrie

Evidence of a palimpsest in K1 and K2 discoverable via steganographic transforms of the Kryptos cipher and plain texts.

By

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Contents

Introduction.....	3
Abstract.....	4
Part I	5
Background	5
Interventions post decryption of K1-K3	6
Issues with the decryption of K2	6
The Berlin Clue	6
The Sculpture Text	7
Complete Cipher Text and Decryption Tableau as it appears on the sculpture.	7
Part II	8
Decryption Steps And Their Derivation	8
The discovery of a null cipher in K1.	8
First Steganographic Transform	11
Second Steganographic Transform.....	13
Derivation of a start point	14
Row Address Scheme	18
Part III.....	19
Potential Masking and Substitution Schemes	19
Possible Masking Scheme for rows proceeding left to right	20
Possible Masking Scheme for rows proceeding right to left	26
Conclusion	28
Appendices.....	29
Appendix A.	29
Elemental Substitution scheme.....	29
Appendix B.	30
Embedded Morse?	30
Appendix C.	30
Algorithms used.....	30

Introduction

The following paper sets out a series of what are assumed to be novel discoveries in relation to the remaining unsolved cipher texts embedded in Jim Sanborn's Kryptos and Antipodes sculptures.

The readership is aimed primarily at those in the Kryptos K4 community who are already conversant with the cipher and plain texts of sections K1-K3 as well as the associated tools and techniques. Such readers may wish to skip the background notes and proceed directly to the later paragraphs.

As with sections K1, K2 and K3 anyone wishing to repeat these results for themselves needs only the decryption tableau, a pen and some paper (although for reasons of pragmatism a computer of some description may allow for faster verification of techniques and results).

The paper puts forward the idea that K4 per se is in fact a palimpsest of K1 and K2 that is discoverable via a steganographic transformation of the underlying cipher text. It is further hypothesised that the resulting palimpsest has had masking and substitution applied prior to final decryption using the Vigenère method with the formally recognised K4 cipher as its key. The paper has been informally titled Valkyrie in homage to the 20th of July plot of 1944, the reasoning being that just like the plot, its credibility will ultimately rest on whether or not it can assume credible control over Berlin (*the known plain text in positions 64-69 of K4*).

I'm indebted to all the hard work undertaken in both the recognized forums and micro sites that have sprung up along the way in support of the commonly shared goal and hope that in time this work finds a place in the canon of standard references cited in the final and complete decryption.

This introduction would be incomplete without a sincere expression of gratitude to Jim Sanborn, Ed Scheidt and their Federal patrons for seeding the world with such a constructive curiosity.

Lastly I'd like to thank Elonka Dunin who took the time out to cast an eye over my initial paper back in 2010. Anyone unfamiliar with Kryptos may also want to visit her Kryptos site at <http://www.elonka.com/kryptos> before proceeding any further with this document.

Apologies are given in advance for anyone with colour blindness as the colour palettes used to render the graphics contained within this document may cause confusion. Please feel free to make any necessary adjustments to the colours if reproducing this work for wider publication.

Abstract

This paper will set out the method by which the 6 letter string BERLIN can be decrypted to plain text using the string NYPVTT in positions 64-69 of K4. It will be shown that this can be achieved by steganographic manipulation of both the cipher and known plain texts into the form given here.

USDJKZLDRNSHGNFIVJ	EMUFPHZLRFAXY	SHADINGANDTHEABSENC	BETWEENSUBTLE
YQTQUXQBQVYUULLTREV	JYQTMKYRDMFD	EOFLIGHTLIESTHENUAN	CEOFIQLUSION
VFPJUDEEHZWETZYVGWH	KKQETGFQJNCE	ITWASTOTALLYINVISIB	LEHOWSTHATPO
GGWHKK?DQMC PFQZDQMM	IAGPFXHQRLG	SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA	ICFIELDXTHEINFORMAT	IONWASGATHER
QZGZLECGYUXUEENJTB	LBQCRTBJDFHRR	EDANDTRANSMITTEDUND	ERGRUUNDTOANU
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI	NKNOWNLOCATIONXDOES	LANGLEYKNOWA
HHDDDUVH?DWKBFUFPN	TDFIYCUQZERE	BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
EVLDKFEZMOQQJLTUGS	YQPFEUNLAVIDX	THERESOMEWHEREXWHOK	NOWSTHEEXACTL
FLGGTEZ?FKZBSFDQVGO	GIPUFXXHHRKF	OCATION?ONLYWWTHISW	ASHISLASTMES
FHQNTGPUAECNUVPDJMQ	CLQUMUNEDFQ	SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE	VENMINUTESSIXPOINTF	IVESECONDSNO
DNQFMPNZGLFLPMRJQYA	LMGNUMVPDXVKP	RTHSEVENTYSEVENDEGR	EESSEIGHTMINU
DQUMEBEDMDHDAFMJGZNU	PLGEWJLLAETGO	TESFORTYFOURSECONDS	WESTXLAYERTWO
ENDYAHROHNLSRHEOCP	EOIBIDYSHNAIA	SLOWLYDESPARATLYSLO	WLYTHEREMAINS
CHTNREYULDSLSSLNOH	SNOSMRWXMNE	OPASSAGEDEBRISTHAT	ENCUMBEREDT
TPRNGATIHNRRARPESLNN	ELEBLPIIACAE	HELOWERPARTOFTHEDOO	RWAYWASREMOV
WMTWNDITEENRAHCTENE	UDRETNHAEOE	EDWITH TREMBLINGHAND	SIMADEATINY
TFOLSEDTIWENHAEIOYT	EYQHEENCTAYCR	BREACHINTHEUPPERLEF	THANDCORNERAN
EIFTBRSPAMHHEWENATA	MATEGYEERLB	DTHENWIDENINGTHEHOL	EALITTLEIIN
TEEFOASFIOTUETUAET	OARMAEERTNRTI	SERTEDTHECANDLEANDP	EEREDINTHEHOT
BSEDDNIAAHTTMSTEWPI	EROAGRIEWFEB	AIRESCAPINGFROMTHEC	HAMBERCAUSED
AECTDDHILCEIHSITEGO	EAOSDDRYDLORIT	THEFLAMETOFLICKERBU	TPRESENTLYDETA
RKLMLEHAGTDHARDPNEO	HMGFMFEUHE	ILSOFTHEROOMWITHINE	MERGEDFROM
ECDMRI PF EIMEHNLSSTT	RTVDOHW?OBKR	THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
UOXOGHULBSOLIFBBWFL	RVQQPRNGKSSO	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXX
TWTQSJQSSEKZZWATJKL	UDIAWINFBNYP	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXBER
VTTMZFPKWGDKZXTJCDI	GKUHUAUEKCAR	LINXXXXXXXXXXXXXXXXX	XXXXXXXXXXXX

The reasoning for making this transform will be shown from first principles, which will then lead directly to a second steganographic transform that will lay bare the palimpsest that underlies the encryption scheme of K4. The existence of a demonstrable row scheme for K4 should significantly decrease the time required to make a final and complete decryption. The paper concludes with a number of suggestions for the masking and substitution methods required for a complete decryption.

Part I

Background

The following information on the background of the Kryptos sculpture was gathered from a range of internet sources.

Kryptos was dedicated on November the 3rd 1990 on the grounds of the CIA headquarters in Langley Virginia. The main body of the sculpture is a worked copper scroll by Artist Jim Sanborn who was commissioned by the CIA to create the installation for the courtyard of their headquarters. He was additionally loaned the services of a retiring CIA employee by the name of Ed Scheidt who was on hand to provide assistance with the cryptographic elements of the piece.

Since 1990 there have been three public acknowledgments in relation to the decryption of parts K1, K2 and K3.

A civilian computer scientist named Jim Gillogly went public with a decryption of parts K1, K2 and K3 in 1999 having used computational analysis to arrive at the plain texts.

After this announcement it was revealed that a CIA employee named David Stein was recognised within the CIA as having cracked parts K1, K2 and K3 using a pencil and paper technique in 1998.

It was later announced in 2005 that an NSA team led by Ken Miller, Dennis McDaniels and two other unnamed individuals had deciphered parts K1, K2 & K3 in late 1992 using computational analysis.

In the years since Gillogly's announcement a mystique has continued to grow around the unsolved K4 cipher text. This is partly due to its resistance to yield its final secrets, but also the mass of interest aroused worldwide by Dan Brown's code sleuthing novels and the growing pervasiveness of cryptography in modern life. As a result there has been a mass of internet based collaboration in the public domain ever since, whereby common interest groups and individuals have persistently poured over the clues left in the plain texts of parts K1, K2 and K3 in search of possible routes into the plain text of K4.

For anyone not familiar with plain texts of K1, K2 and K3 they read as follows.

K 1 (this was encrypted using the Keyword PALIMPSEST)

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF
IQLUSION

K2 (this was encrypted using the Keyword ABSCISSA) *Note: This was publicly corrected by Jim Sanborn on April 19th, 2006 who intervened to point out the IDBYROWS should read XLAYERTWO*

IT WAS TOTALLY INVISIBLE HOWS THAT POSSIBLE? THEY USED THE EARTHS
MAGNETIC FIELD X THE INFORMATION WAS GATHERED AND TRANSMITTED
UNDERGRUUND TO AN UNKNOWN LOCATION X DOES LANGLEY KNOW ABOUT THIS?
THEY SHOULD ITS BURIED OUT THERE SOMEWHERE X WHO KNOWS THE EXACT
LOCATION? ONLY WW THIS WAS HIS LAST MESSAGE X THIRTY EIGHT DEGREES
FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY SEVEN DEGREES
EIGHT MINUTES FORTY FOUR SECONDS WEST X LAYER TWO

K 3 (this section was not encrypted using the Kryptos Tableau, but instead made use of a 192 character skip sequence to encrypt the plain text)

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT
ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH
TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND
THEN WIDENING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED IN THE
HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT
PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU
SEE ANYTHING Q

Interventions post decryption of K1-K3

Issues with the decryption of K2

With regards the correct decryption of K2, Jim Sanborn the artist that created Kryptos personally intervened on April 19th 2006, contacting the group moderators in order to say, "No, that last part is wrong." Elonka's Kryptos website at www.elonka.com/kryptos relays the following about this incident.

"He also indicated that there was a missing character on the sculpture, probably something that would have resulted in a plaintext "X" before that section. He said that he had thought that with the missing character, the section in question would have come out to be an unintelligible scramble. Instead, he was astounded to see that by sheer chance, the resulting random text had turned out to be apparently intelligible English, "ID BY ROWS", although that was not what was intended."

The Berlin Clue

On the 21st November 2010 Jim Sanborn once again intervened, but in this instance it was to reveal a clue. Sanborn indicated that a six letter string between positions 64 through to 69 of the Kryptos plaintext for K4 would read BERLIN once decrypted.

The Sculpture Text

The text as it appears on the sculpture has been lifted wholesale from Elonka's website and is presented in full below.

Complete Cipher Text and Decryption Tableau as it appears on the sculpture.

EMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ YQTQUXQBQVYUUVLLTREVJYQTMKYRDMFD VFPJUDEEHZWETZYVGWHKKQETGFQJNCE GGWHKK?DQMCPPFQZDQMMIAGPFXHQRLG TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA QZGZLECGYUXUEENJTBQLBQCRTBJDFHRR YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE EVLDKFEZMOQQJLTTUGSYQPFEUNLAVIDX FLGGTEZ?FKZBSFDQVGOGIPUFXXHDKRF FHQNTGPAECNUVPDJMQCLQUMUNEDFQ ELZZVRRGKFFVOEEXBDMVVPNFQXZLGRE DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG	ABCDEFGHIJKLMNOPQRSTUVWXYZABCD AKRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT BRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT CYPTOSABCDEFGHIJLMNQUVWXXZKRYPTO DPTOSABCDEFGHIJLMNQUVWXXZKRYPTOS ETOSABCDEFGHIJLMNQUVWXXZKRYPTOSA FOSABCDEFGHIJLMNQUVWXXZKRYPTOSAB GSABCDEFGHIJLMNQUVWXXZKRYPTOSABC HABCDEFGHIJLMNQUVWXXZKRYPTOSABCD IBCDEFGHIJLMNQUVWXXZKRYPTOSABCDE JCDEFGHIJLMNQUVWXXZKRYPTOSABCDEF KDEFGHIJLMNQUVWXXZKRYPTOSABCDEFG LEFGHIJLMNQUVWXXZKRYPTOSABCDEFGH MFGHIJLMNQUVWXXZKRYPTOSABCDEFGHI
ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIA CHTNREYULDSLLSLLNOHSNOSMRWXMNE TPRNGATIHNRARPESLNNELEBLPIIACAE WMTWNDITEENRAHCTENEUDRETNHAEOE TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR EIFTBRSPAMHHEWENATAMATEGYEERLB TEEFOASFIOTUETUAEOTOARMAAEERTNRTI BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB AECTDDHILCEIHSITEGOEAOSSDDRYDLORIT RKLMLHAGTDHARDPNEOHMGFMFEUHE ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO TWTQSJSSEKZZWATJKLUDIAWINFBNYP VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR	NGHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL OHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL PIJLMNQUVWXXZKRYPTOSABCDEFGHIJLM QJLMNQUVWXXZKRYPTOSABCDEFGHIJLMN RLMNQUVWXXZKRYPTOSABCDEFGHIJLMNQ SMNQUVWXXZKRYPTOSABCDEFGHIJLMNQU TNQUVWXXZKRYPTOSABCDEFGHIJLMNQUV UQUVWXXZKRYPTOSABCDEFGHIJLMNQUVW VUVWXXZKRYPTOSABCDEFGHIJLMNQUVWX WVWXXZKRYPTOSABCDEFGHIJLMNQUVWXZ XWXXZKRYPTOSABCDEFGHIJLMNQUVWXZK YXZKRYPTOSABCDEFGHIJLMNQUVWXZKR ZZKRYPTOSABCDEFGHIJLMNQUVWXZKRY ABCDEFGHIJKLMNOPQRSTUVWXYZABCD

K1 is highlighted in Mauve, K2 in Blue, K3 in Green and K4 in Red

Part II

Decryption Steps And Their Derivation

Before commencing this section it should be noted that K4 is not a 'block cipher' that can be decrypted by a single key. It will become evident throughout each step that the whole of the Kryptos sculpture represents a holistic puzzle that is only 'eventually' decrypted through use of the known K4 string. There are many facets to the puzzle, but viewed as a whole it will be seen that the aim of the piece is to quite literally step us through the art and history of cryptography itself. This is done quite brilliantly and bears testament to the intellect and artistry of both Scheidt and Sanborn. I suspect that a further study of the cipher and plain texts after this paper has been distributed will yield many more encryption methods, some of which may light a way to a decryption path that does not require foreknowledge of the BERLIN clue.

The discovery of a null cipher in K1.

I have included the following working details for extraction of the null cipher in K1 for the purposes of completeness. Readers from outside the Kryptos community that are not looking to reproduce the results may safely skip to the next heading which is 'First Steganographic Transform' on page 11.

The presence of a null cipher in K1 is found by following and solving the clues given in the plain texts of K1 and K3.

The first clue was found in the first half of the K1 text.

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT

If K1 is treated as something akin to a cryptic crossword clue then the following reasoning can be applied.

i) Compare between '*subtle shading and absence of*' (light) by writing one over the other:

SUBTLE SHADING
ABSENCE OF

ii) Remove the letters that make up LIGHT from SUBTLE SHADING to give:

SUBESADN
ABSENCEOF

iii) Remove common letters in each word to leave:

USD
CEOF

USD is present in the cipher text of K1 whilst CEOF is present in the plain text.

EMUFPHZLRFAXY USD KZLDRNSHG NFIVJYQTQUXQBQVYUVLLTREVJYQTMKYRDMFD

BETWEENSUBTLESHADINGANDTHEABSENCEOFLIGHTLIESTHENUAN CEOF IOLUSION

Significance of the extracted information

It is hypothesised that USD and CEOF are an example of 'Keys', the idea being that once the cryptographer has extracted the correct key from a given clue that they are then able to unlock otherwise hidden information from within the rest of the sculpture texts.

To substantiate this hypothesis it will now be shown how the strings USD and CEOF in K1 can be used to find the entry point into the decryption process.

To begin, let us first temporarily discard the first 13 characters, in order to give a start point of USD.

USDJKZLDRNSHGNFIVJYQTQUXQBQVYUUVLLTREVJYQTMKYRDMFD

Using the Vigenère scheme to decrypt this string using K4 leads to a small but significant observation. The 'encryption' rather than decryption begins with a K. The idea that we might have to encrypt rather than decrypt is interesting as it is our first encounter with the idea that symmetry may play a part in the decryption process. Given that we had to decrypt in order to arrive at the plain texts of K1 and K2, it seems counter intuitive at first that we may then have to do the inverse of this process.

Some pointers appear to be given in the form of intentional spelling mistakes, that we should consider symmetry as being an important aspect of the decryption process. The spelling mistake AU has been commented on many times, but if we consider it to mean Gold then we know its atomic number is 79. This is obviously a mirror image of 97 (the number of characters in K4), which may indicate that we should be thinking about the role of symmetry when dealing with the sculpture holistically. A number of other clues point to symmetry including the water feature as well as Sanborn's decision to create the sculpture in the form of two semi-circles rather than as a cylinder.

By referring back to the string CEOF it is possible to cite two other clues as having relevance to the next step. The first of these clues is from one of the Morse Code messages left dotted around the sculpture. Once resolved one particular Morse message reads, 'SHADOW FORCES'.

If we quite literally shade the word FORCES then we see the recurring relation of CEOF. Admittedly the R and the S is harder to justify, but not everything is given to us on a plate as we shall see. If we now consider K1's enigmatic message as a whole we are reminded that 'BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF IQLUSION'. From the perspective of rendering something graphically 'occlusion' is the act of blocking. There is another aspect of CE OF that is interest. CF is often used as an abbreviation for 'compare'. By placing CE next to OF we are in fact comparing and finding that CF can be fashioned by occluding both two letter strings. The eventual significance of this string will be found in the concluding part of the paper.

This leaves us with the strong indication that O and C as well as E and F may share an occlusion relationship when shaded. With this in mind let us examine the first four letters of K4. These are found to be OBKR. By thinking about possible occlusion relationships we can transform OBKR to CRYP. Returning to our previous observation that the encryption of (the reduced form starting at USD) of K4 gives us an encrypted string starting with a K we can make a deductive leap and reason that the target we should be looking to match is in fact KRYP.

The string required to give us an encrypted cipher text of KRYP is UNYY. If we now search for such a string in the K1 cipher text then we make our first significant discovery.

USDJKZLDRNSHGNFIVJYQTQUXQBQVYUUVLLTREVJYQTMKYRDMFD

The length between letters proceeds in the pattern 10, 9, 10. On its own this relationship might merely be dismissed as a coincidence, but its presence is cited a second time in K3.

The line in question reads: I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER

Extending the pattern backwards in both K1 and a cyclically repeating K4 string of ROBKR yields:

EMUF^PHIZLRFAXY^USDJKZLDK^RNSHG^NFNIV^JYQTQUXQBQ^VYUVLLTREVJYQTMKYRDMFD

PUNYY encrypted using a cipher key of ROBKR gives TKRYP.

PUNY is of course a synonym of TINY. Our first letter in this scheme is T which may relate to another Morse message that reads: T IS YOUR POSITION.

The length between letters now proceeds in the pattern 9,10,9,10. Adding these numbers together yields a value of 38. Adding an additional sequence of 9,10 would of course give us 57. The fact that this particular skip sequence was capable of generating the first two coordinates or 38 and 57 degrees given in K2 gave a strong indication that this was a significant step in the decryption scheme.

The most fascinating part of this particular component of the puzzle is that there appears to be an almost explicit reference as to what it represents. Using a symbol font in MS Word and typing 'hull' allows us to create a mirror image of the lower case string in Greek.

Original Key	unyy
Key as typed to achieve Greek mirror text	hull
Mirror Key	ηυλλ

By occluding the h to give an n, we actually have the word 'null' right in front of us. By treating UNYY as a null cipher we end up with another interesting relationship, but this time in the corresponding plain text. I shall also include the first line of K2. Remember that the entry point was arrived at via UNYY not PUNYY so in this illustration you will notice the first 13 letters are discarded. I shall circle the letters corresponding to UNYY in the plaintext.

SHADINGANDTHEABSENC

EOFLIGHTLIESTHENUANCEOFIQLUSION

ITWASTOTALLYINVISIBLEHOWSTHATPO

It can be seen that laying out the lettering in this mode leads to the appearance of an interesting letter string in the vertical plane.

SHADINGANDTHEABSENC
EOFLIGHTLIESTHENUANCEOFIQLUSION
ITWASTOTALLYINVISIBLEHOWSTHATPO

The plain text of K3 that reads HOT AIR ESCAPING is immediately brought back to our immediate attention. The null cipher has now given way to what appears to be some kind of running key that only exposes itself upon structural change to the text layout.

First Steganographic Transform

Taking our cue from the keys derived above and reordering the Cipher and Plain Texts now gives the following layout. It will be noted that the first 13 characters have now been inserted after the string USDJKZLDKRNSHGNFIVJ. I shall highlight the positions of USD and CEOF as well as the location of two strings in the vertical plane that are of interest. It should also be pointed out that the XY string combo now sits in a position that is consistent with the coordinates of a graph and resonates with the Cartesian aspects of the Keyword ABSCISSA used of the decryption of K2.

USDJKZLDKRNSHGNFIVJEMUFPHZLRFAXY YQTQUXQBQVYUVLITREVJYQTMKYRDMFD VFPJUDEEHZWETZYVGWHKKQETGFQJNCE GGWHKK?DQMC PFQZDQMMIAGPFXHQRLG TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA QZGZLECGYUXUEENJTBJLBQCRBTBJDFHRR YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE EVLDKFEZMOQQJLTUGSYQPFEUNLAVIDX FLGGTEZ?FKZBSFDQVGOGIPUFXXHDKRF FHQNTG PUAECNUVPDJMQCLQUMUNEDFQ ELZZVRRGKFFVOEEXBDMVPNFQXEZLGRE DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP DQUMEBEDMHDAFMJGZNUPLGEWJLLAETGO ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIA CHTNREYULDSL LLLNOHSNOSMRWXMNE TPRNGATIHNRRARPESLNNELEBLPIIACAE WMTWNDITEENRAHCTENEUDRETNHAE OE TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR EIFTBRSPAMHHEWENATAMATEGYEERLB TEEFOASFIOTUETUAEO TOARMAEERTNRTI BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB AECTDDHILCEIHSITEGOEAO SDDRYDLORIT RKLMLHAGTDHARDPNEOHMGFMFEUHE ECDMRI PF EIMEHNLSSTTRTVDOHW?OBKR UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO TWTQSQSSEKZZWATJKLUDIAWINFBNYP VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR	SHADINGANDTHEABSENCBETWEENSUBTLE E OFLIGHTLIESTHENUAN CEOF IQLUSION ITWASTOTALLYINVISIBLEHOWSTHATPO SSIBLE?THEYUSEDTHEEARTHSMAGNET ICFIELDXTHEINFORMATIONWASGATHER EDANDTRANSMITTEDUNDERGRUUNDTOANU NKNOWNLOCATIONXDOESLANGLEYKNOWA BOUTTHIS?THEYSHOULDI TSBURIEDOUT THERESOMEWHEREXWHOKNOWSTHEEXACTL OCATION?ONLYWWTHISWASHISLASTMES SAGEXTHIRTYEIGHTDEGREESFIFTYSE VENMINUTES SIXPOINTFIVESECONDSNO RTHSEVENTYSEVENDEGREESEIGHTMINU TESFORTYFOURSECONDSWESTXLAYERTWO SLOWLYDESPARATLYSLOWLYTHEREMAINS OFPASSAGEDEBRISTHATENCUMBEREDT HELOWERPARTOFTHE DOORWAYWASREMOV EDWITH TREMBLINGHANDS IMADEATINY BREACHINTHEUPPERLEFTHANDCORNERAN DTHENWIDENINGTHEHOLEALITTLEI IN SERTEDTHECANDLEANDPEEREDINTHEHOT AIRES CAPINGFROMTHECHAMBERCAUSED THEFLAMETOFLICKERBUTPRESENTLYDETA ILSOFTHEROOMWITHINEMERGEDFROM THEMISTXCANYOUSEEANYTHINGQ?XXXX VERXXXXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXXXBER LINXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
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The next part of the transform is to examine the position of CEOF and recognise its significance as a boundary marker. If we split the text upon reaching this string from left to right then we notice that the left hand division has rows that follow the skip pattern outlined in the null cipher of 10,9.

USD JKZLDKRNSHGNFIVJ	EMUFPHZLRFAXY	SHADINGANDTHEABSENC	BETWEENSUBTLE
YQTQUXQBQVYUULLTREV	JYQTMKYRDMFD	EOFLIGHTLIESTHENUAN	CEO FJQLUSION
VFPJUDEEHZWETZYVGWH	KKQETGFQJNCE	ITWASTOTALLYINVISIB	LEHOWSTHATPO
GGWHKK?DQMC PFQZDQMM	IAGPFXHQRLG	SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA	ICFIELDXTHEINFORMAT	IONWASGATHER
QZGZLECGYUXUEENJTBJ	LBQCRTBJDFHRR	EDANDTRANSMITTEDUND	ERGRUUNDTOANU
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI	NKNOWNLOCATIONXDOES	LANGLEYKNOWA
HHDDDUVH?DWKBFUF PWN	TDFIYCUQZERE	BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
EVLDKFEZMOQQJLTTUGS	YQPFEUNLAVIDX	THERESOMEWHEREXWHOK	NOWSTHEEXACTL
FLGGTEZ?FKZBSFDQVGO	GIPUFXXHHRKF	OCATION?ONLYWWTHISW	ASHISLASTMES
FHQNTG PUAECNUVPDJMQ	CLQUMUNEDFQ	SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE	VENMINUTESSIXPOINTF	IVESECONDSNO
DNQFMPNZGLFLPMRJQYA	LMGNUMPD XVKP	RTHSEVENTYSEVENDEGR	ESEIGHTMINU
DQUMEBEDMHDAFMJGZNU	PLGEWJLLAETGO	TESFORTYFOURSECONDS	WESTXLAYERTWO
ENDYAHROHNLSRHEOCPT	EOIBIDYSHNAIA	SLOWLYDESPARATLYSLO	WLYTHEREMAINS
CHTNREYULDSL LSL LNOH	SNOSMRWXMNE	OPASSAGEDEBRISTHAT	ENCUMBEREDT
TPRNGATIHNRRARPESLNN	ELEBLPIIACAE	HELOWERPARTOFTHEDOO	RWAYWASREMOV
WMTWNDITEENRAHCTENE	UDRETNHAE OE	EDWITH TREMBLINGHAND	SIMADEATINY
TFOLSEDTIWENHAEIOYT	EYQHEENCTAYCR	BREACHINTHEUPPERLEF	THANDCORNERAN
EIFTBRSPAMHHEWENATA	MATEGYEERLB	DTHENWIDENINGTHEHOL	EALITTLEIIN
TEEFOASFIOTUETUAEOT	OARMAEERTNRTI	SERTEDTHECANDLEANDP	EEREDINTHEHOT
BSEDDNIAAHTTMSTEWPI	EROAGRIEWFE B	AIRESCAPINGFROMTHEC	HAMBERCAUSED
AECTDDHILCEIHSITEGO	EAOSDDRYDLORIT	THEFLAMETOFLICKERBU	TPRESENTLYDETA
RKLMLEHAGTDHARDPNEO	HMGFMFEUHE	ILSOFTHEROOMWITHINE	MERGEDFROM
ECDMRI PF EIMEHNLSSTT	RTVDOHW?OBKR	THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
UOXOGHULBSOLIFBBWFL	RVQQPRNGKSSO	VERXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXXX
TWTQJSQSSEKZZWATJKL	UDIAWINFBNYP	XXXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXBER
VTTMZFPKWGDKZXTJCDI	GKUHUAUEKCAR	LINXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXXX

The most striking feature of this particular layout is that it almost looks like a facsimile of the original sculpture text. Once laid out in this format we can now proceed to the most significant discovery, which is the existence of further embedded steganographic information within the cipher and plain text. Before moving on to this step it is worth pointing out the clues that lead us to such a radical conclusion. It has been pointed out by others before that the hours and minutes of the coordinates given in K2 sum to 180. To recap this calculation and make things explicit we can see that:

$$38 + 57 + 77 + 8 = 180.$$

This is an interesting piece of math and once again points in the direction of symmetry having something to do with the final decryption of K4. The first real indication we have that this summation of the coordinates is significant is in the form of an anagram embedded within the first line of K3. I have drawn attention to the original text above, but it can be seen by re-arrangement that the anagram reads as: WESTHLAYERMIN. Again our attention is being drawn to the coordinates. Their relevance in the decryption scheme will now be demonstrated.

Second Steganographic Transform

First let us remind ourselves of the coordinate scheme given in the K2 plaintext.

THIRTY EIGHT DEGREES FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH
SEVENTY SEVEN DEGREES EIGHT MINUTES FORTY FOUR SECONDS WEST

The next steganographic transform relates directly to this clue and can best be explained by breaking the text up into separate blocks of colour. I apologise in advance for the English spelling as well as any difficulties in discerning the colours, but please feel free to render this as you wish when reproducing these results.

USDJKZLDRNSHGNFIVJ	EMUFPHZLRFAXY	SHADINGANDTHEABSENC	BETWEENSUBTLE
YQTQUXQBQVYUULLTREV	JYQTMKYRDMFD	EOFLIGHTLIESTHENUAN	CEOFIQLUSION
VFPJUDEEHZWETZYVGWH	KKQETGFQJNCE	ITWASTOTALLYINVISIB	LEHOWSTHATPO
GGWHKK?DQMCPFQZDQMM	IAGPFXHQRLG	SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA	ICFIELDXTHEINFORMAT	IONWASGATHER
QZGZLECGYUXUEENJTB	LBQCRTBJDFHRR	EDANDTRANSMITTEDUND	ERGRUUNDTOANU
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI	NKNOWNLOCATIONXDOES	LANGLEYKNOWA
HHDDDUVH?DWKBFUFPNW	TDFIYCUQZERE	BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
EVLDKFEZMOQQJLTTUGS	YQPFEUNLAVIDX	THERESOMEWHEREXWHOK	NOWSTHEEXACTL
FLGGTEZ?FKZBSFDQVGO	GIPUFXXHHRKF	OCATION?ONLYWWTHISW	ASHISLASTMES
FHQNTGPPUAECNUVPDJMQ	CLQUMUNEDFQ	SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE	VENMINUTESSIXPOINTF	IVESECONDSNO
DNQFMPNZGLFLPMRJQYA	LMGNUMVPDXVKP	RTHSEVENTYSEVENDEGR	EESSEIGHTMINU
DQUMEBEDMDHDAFMJGZNU	PLGEWJLLAETGO	TESFORTYFOURSECONDS	WESTXLAYER TWO
ENDYAHROHNLSRHEOCP	EOIBIDYSHNAIA	SLOWLYDESPARATLYSLO	WESTHLAYERMIN
CHTNREYULDSLSSLNOH	SNOSMRWXMNE	OPPASSAGEDEBRISTHAT	ENCUMBEREDT
TPRNGATIHNRRARPESLNN	ELEBLPIIACAE	HELOWERPARTOFTHEDOO	RWAYWASREMOV
WMTWNDITEENRAHCTENE	UDRETNHAEOE	EDWITH TREMBLINGHAND	SIMADEATINY
TFOLSEDTIWENHAEIOYT	EYQHEENCTAYCR	BREACHINTHEUPPERLEF	THANDCORNERAN
EIFTBRSPAMHHEWENATA	MATEGYEERLB	DTHENWIDENINGTHEHOL	EALITTLEIIN
TEEFOASFIOTUETUAEO	OARMAEERTNRTI	SERTEDTHECANDLEANDP	EEREDINTHEHOT
BSEDDNIAAHTTMSTEWPI	EROAGRIEWFEB	AIRESCAPINGFROMTHEC	HAMBERCAUSED
AECTDDHILCEIHSITEGO	EAOSDDRYDLORIT	THEFLAMETOFLICKERBU	TPRESENTLYDETA
RKLMLEHAGTDHARDPNEO	HMGFMFEUHE	ILSOFTHEROOMWITHINE	MERGEDFROM
ECDMRI PFEIMEHNLSSTT	RTVDOHW?OBKR	THEMISTXCANYOUSEAN	YTHINGQ?
UOXOGHULBSOLIFBBWFL	RVQQPRNGKSSO		
TWTQSQJSSEKZZWATJKL	UDIAWINFBNYP		
VTTMZFPKWGDKZXTJCDI	GKUHUAUEKCAR		

Mappings to H and MIN of K2 Coordinate System	
38 Characters	
19 Characters	38 + 19 = 57
6(?)5 Characters	
77 Characters	

If the question mark is equated to a decimal point then we can see that it is possible to derive the coordinates 38, 57, 6.5 and 77. Other coordinates appear to be alluded to, but these will be ignored for the purposes of explaining the role that the central block of 77 characters plays in the decryption process. Please extend on this work and explore the possible meanings of the other coordinates as I'm sure the cipher and plain texts still have plenty of secrets to impart.

Derivation of a start point

The derivation of the start point for the decryption was stumbled upon by pure chance, but it will be seen how this start point complements the hidden row mechanism established in the previous section and leads us towards a credible decryption process. Before we do this let us make another bold leap of faith and guess that the cipher and plain texts were designed to be symmetrical. This would mean that the block coordinate groupings found in the last section could be mirrored in the X axis running between the edge of K2 and K3. Such a scheme yields the following layout. You will note that the anagram in the top right of K3 has been rearranged in all the plain texts from this point in.

USDJKZLDKRNSHGNFIVJ	EMUFPHZLRFAXY	SHADINGANDTHEABSENC	BETWEENSUBTLE
YQTQUXQBQVYUVLLTREV	JYQTMKYRDMFD	EOFLIGHTLIESTHENUAN	CEOFIQLUSION
VFPJUDEEHZWETZYVGWH	KKQETGFGJNCE	ITWASTOTALLYINVISIB	LEHOWSTHATPO
GGWHKK?DQMCFQZDQMM	IAGPFXHQRLG	SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA	ICFIELDXTHEINFORMAT	IONWASGATHER
QZGZLECGYUXUEENJTB	LBQCRTBJDFHRR	EDANDTRANSMITTEDUND	ERGRUUNDTOANU
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI	NKNOWNLOCATIONXDOES	LANGLEYKNOWA
HHDDDUVH?DWKBFUFPWN	TDFIYCUQZERE	BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
EVLDKFEZMOQQJLTTUGS	YQPFEUNLAVIDX	THERESOMEWHEREXWHOK	NOWSTHEEXACTL
FLGGTEZ?FKZBSFDQVGO	GIPUFXHHDRKF	OCATION?ONLYWWTHISW	ASHISLASTMES
FHQNTGPUAECNUVPDJMQ	CLQUMUNEDFQ	SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE	VENMINUTESIXPOINTF	IVESECONDSNO
DNQFMPNZGLFLPMRJQYA	LMGNUMVPDXVKP	RTHSEVENTYSEVENDEGR	EESIEIGHTMINU
DQUMEBEDMHDAFMJGZNU	PLGEWJLLAETGO	TESFORTYFOURSECONDS	WESTXLAYERTWO
ENDYAHROHNLSRHEOCP	EOIBIDYSHNAIA	SLOWLYDESPARATLYSLO	WESTHAYERMIN
CHTNREYULDSLILLSLNOH	SNOSMRWXMNE	OPASSAGEDEBRISTHAT	ENCUMBEREDT
TPRNGATIHNRRARPESLNN	ELEBLPIIACAE	HELOWERPARTOFTHEDOO	RWAYWASREMOV
WMTWNDITEENRAHCTENE	UDRETNAHAE	EDWITHTREMBLINGHAND	SIMADEATINY
TFOLSEDTIWENHAEIOYT	EYQHEENCTAYCR	BREACHINTHEUPPERLEF	THANDCORNERAN
EIFTBRSPAMHHEWENATA	MATEGYEERLB	DTHENWIDENINGTHEHOL	EALITTLEIIN
TEEFOASF IOTUETUA EOT	OARMAEERTNRTI	SERTEDTHECANDLEANDP	EEREDINTHEHOT
BSEDDNIAAHTTMSTEWPI	EROAGRIEWFEB	AIRESCAPINGFROMTHEC	HAMBERCAUSED
AECTDDHILCEIHSITEGO	EAOSDDRYDLORIT	THEFLAMETOFLICKERBU	TPRESENTLYDETA
RKLMLEHAGTDHARDPNEO	HMGFMFEUHE	ILSOFTHEROOMWITHINE	MERGEDFROM
ECDMRI PFEIMEHNLSTT	RTVDOHW?OBKR	THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
UOXOGHULBSOLIFBBWFL	RVQQPRNGKSSO	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXX
TWTQJSQSSEKZZWATJKL	UDIAWINFBNYP	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXBER
VTTMZFPKWGDKZXTJCDI	GKUHUAEUKCAR	LINXXXXXXXXXXXXXXXX	XXXXXXXXXXXX

We return now to the null cipher that we found in the first section and show its position in the plain text. A rail fence scheme is then shown in K3 that reflects this string. If we take the longer string previously derived of PUNYY then we find the scheme extends backwards along the diagonal.

U S D J K Z L D K E N S H G N F I V J	E M U I P I Z L R F A X Y	S H A D I N G A N D T H E A B S E N C	B E T W E E N S U B T L E
Y Q T Q U X Q B Q V Y U V L L T R E V	J Y Q T M K Y R D M F D	E O F L I G H T L I E S T H E N U A N	C E O F I Q L U S I O N
V F P J U D E E H Z W E T Z Y V G W H	K K Q E T G F Q J N C E	I T W A S T O T A L L Y I N V I S I B	L E H O W S T H A T P O
G G W H K K ? D Q M C P F Q Z D Q M M	I A G P F X H Q R L G	S S I B L E ? T H E Y U S E D T H E E	A R T H S M A G N E T
T I M V M Z J A N Q L V K Q E D A G D	V F R P J U N G E U N A	I C F I E L D X T H E I N F O R M A T	I O N W A S G A T H E R
Q Z G Z L E C G Y U X U E E N J T B J	L B Q C R T B J D F H R R	E D A N D T R A N S M I T T E D U N D	E R G R U N D T O A N U
Y I Z E T K Z E M V D U F K S J H K F	W H K U W Q L S Z F T I	N K N O W N L O C A T I O N X D O E S	L A N G L E Y K N O W A
H H D D D U V H ? D W K B F U F P W N	T D F I Y C U Q Z E R E	B O U T T H I S ? T H E Y S H O U L D	I T S B U R I E D O U T
E V L D K F E Z M O Q Q J L T T U G S	Y Q P F E U N L A V I D X	T H E R E S O M E W H E R E X W H O K	N O W S T H E E X A C T L
F L G G T E Z ? F K Z B S F D Q V G O	G I P U F X H H D R K F	O C A T I O N ? O N L Y W W T H I S W	A S H I S L A S T M E S
F H Q N T G P U A E C N U V P D J M Q	C L Q U M U N E D F Q	S A G E X T H I R T Y E I G H T D E G	R E E S F I F T Y S E
E L Z Z V R R G K F F V O E E X B D M	V P N F Q X E Z L G R E	V E N M I N U T E S S I X P O I N T F	I V E S E C O N D S N O
D N Q F M P N Z G L F L P M R J Q Y A	L M G N U V P D X V K P	R T H S E V E N T Y S E V E N D E G R	E E S E I G H T M I N U
D Q U M E B E D M H D A F M J G Z N U	P L G E W J L L A E T G O	T E S F O R T Y F O U R S E C O N D S	W E S T X L A Y E R T W O
E N D Y A H R O H N L S R H E O C P T	E O I B I D Y S H N A I A	S L O W L Y D E S P A R A T L Y S L O	W E S T H L A Y E R M I N
C H T N R E Y U L D S L L S L L N O H	S N O S M R W X M N E	O F P A S S A G E D E B R I S T H A T	E N C U M B E R E D T
T P R N G A T I H N R A R P E S L N N	E L E B L P I I A C A E	H E L O W E R P A R T O F T H E D O O	R W A Y W A S R E M O V
W M T W N D I T E E N R A H C T E N E	U D R E T N H A E O E	E D W I T H T R E M B L I N G H A N D	S I M A D E A T I N Y
T F O L S E D T I W E N H A E I O Y T	E Y Q H E E N C T A Y C R	B R E A C H I N T H E U P P E R L E F	T H A N D C O R N E R A N
E I F T B R S P A M H H E W E N A T A	M A T E G Y E E R L B	D T H E N W I D E N I N G T H E H O L	E A L I T T L E I I N
T E E F O A S F I O T U E T U A E O T	O A R M A E E R T N R T I	S E R T E D T H E C A N D L E A N D P	E E R E D I N T H E H O T
B S E D D N I A A H T T M S T E W P I	E R O A G R I E W F E B	A I R E S C A P I N G F R O M T H E C	H A M B E R C A U S E D
A E C T D D H I L C E I H S I T E G O	E A O S D D R Y D L O R I T	T H E F L A M E T O F L I C K E R B U	T P R E S E N T L Y D E T A
R K L M L E H A G T D H A R D P N E O	H M G F M F E U H E	I L S O F T H E R O O M W I T H I N E	M E R G E D F R O M
E C D M R I P F E I M E H N L S S T T	R T V D O H W ? O B K R	T H E M I S T X C A N Y O U S E E A N	Y T H I N G Q ? X X X X
U O X O G H U L B S O L I F B B W F L	R V Q Q P R N G K S S O	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXXX
T W T Q S J Q S S E K Z Z W A T J K L	U D I A W I N F B N Y P	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXX B E R
V T T M Z F P K W G D K Z X T J C D I	G K U H U A U E K C A R	L I N X X X X X X X X X X X X X X X X	XXXXXXXXXXXXXX

At this point it was noted that the rail fence scheme could be extended to the top right corner of the mirrored block of 77 characters if the I below SE was included in the vertical plane and then the sequence started again from the H on the right hand side of the initial S. This is best shown graphically as:

(S) (H) A D I N G A N D (T) (H) E A B S E N C B E T W (E)
 (E) (O) F L I G H T L I (E) S T H E N U A N
 (I) T W A S T O T A L L Y I N V I S I B

In order to show this relation in K3 more fully I will zoom in on the plain text to show the complete rail fence scheme alluded to above.

SLOWLYDESPARATLYSIO
 OFPASSAGEDEBRISTHAT
 HELOWERPARTOFTHEDOO
 EDWITHTREMBLINGHAND
 BREACHINTHEUPPERLEF
 DTHENWIDENINGTHEHOL
 SERTEDTHECANDLEANDP
 AIRESCAPINGFROMTHEC
 THEFLAMETOFLICKERBU
 ILSOFTHEROOMWITHINE

With regards to the clue in K2 that reads: THEY USED THE EARTHS MAGNETIC FIELD, we can find the following interesting anagrams along the path of the rail fence.

SLOWLYDESPARATLYSIO
 OFPASSAGEDEBRISTHAT
 HELOWERPARTOFTHEDOO
 EDWITHTREMBLINGHAND
 BREACHINTHEUPPERLEF
 DTHENWIDENINGTHEHOL
 SERTEDTHECANDLEANDP
 AIRESCAPINGFROMTHEC
 THEFLAMETOFLICKERBU
 ILSOFTHEROOMWITHINE

The boxed strings above read off as.

ERHTSA, FLICWT, ETNECG and EGAMEN

EARTHS is definitely extractable from the first anagram, but I'll leave the idea of the presence of MAGNETIC FIELD as an interesting piece of conjecture that others may wish to pursue at their leisure. We will return again to the theme of 'THE EARTHS MAGNETIC FIELD in section III when

we start looking at potential masking schemes, but for now we shall return to our primary task which is to discern the start point for the decryption process.

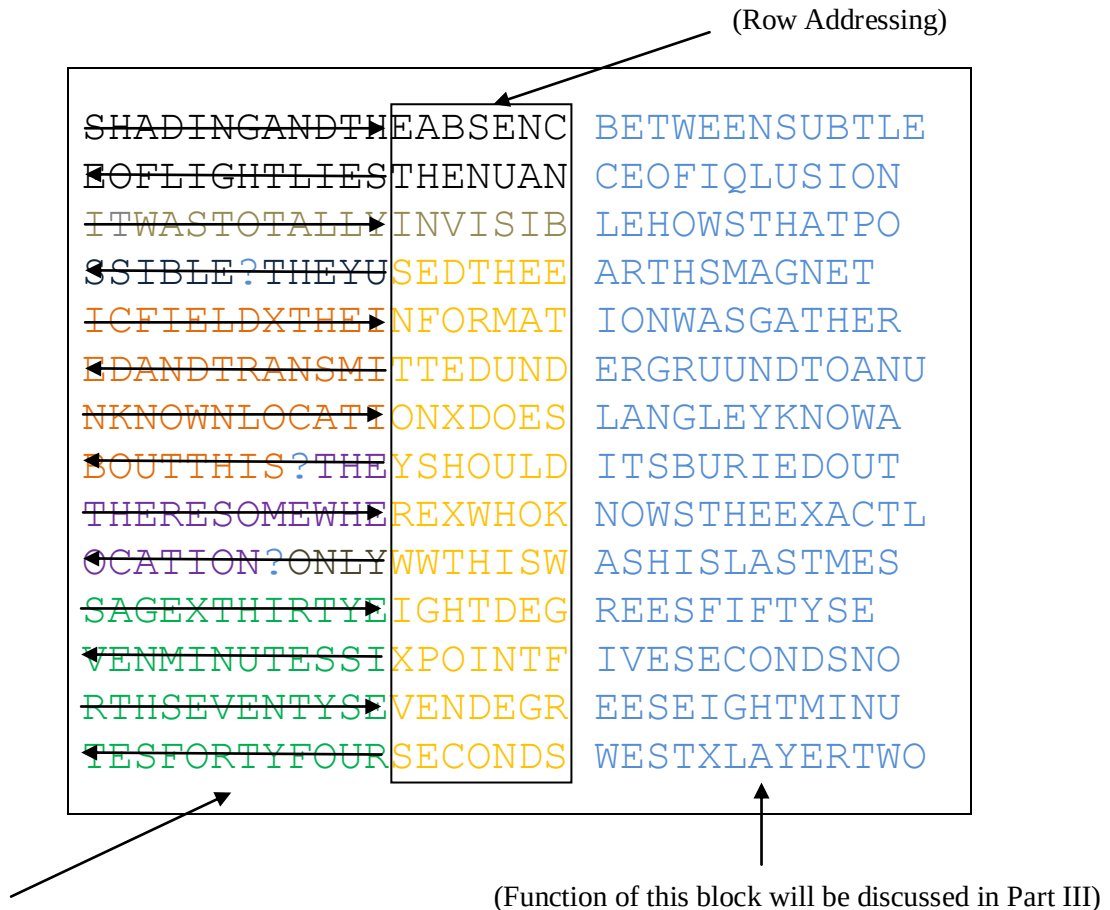
The rail fence scheme can be seen to extend in a diagonal before finally terminating on the O at the top right of the K3 section that has been marked out as being the symmetrical component of the block of 77 characters in K2. Assuming that everything is symmetrical, let's reflect the path of the line in the X axis.

USDJKZLDRNSHGNFIVJ	EMUFPHZLRFAXY	SHADINGANDTHEABSENC	BETWEENSUBTLE
YQTQUXQBQVYUULLTREV	JYQTMKYRDMFD	EOFLIGHTLIESTHENUAN	CEOFIQLUSION
VFPJUDEEHZWETZYVGWH	KKQETGFQJNCE	ITWASTOTALLYINVISIB	LEHOWSTHATPO
GGWHKK?DQMCPFQZDQMM	IAGPFXHQRLG	SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA	ICFIELDXTHEINFORMAT	IONWASGATHER
QZGZLECGYUXUEENJTB	LBQCRTBJDFHRR	EDANDTRANSMITTEDUND	ERGRUUNDOANU
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI	NKNONNLOCATIONXDOES	LANGLEYKNOWA
HHDDDUVH?DWKBFUFPNW	TDFIYCUQZERE	BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
EVLDKFEZMOQQJLTTUGS	YQPFEUNLAVIDX	THERESOMEWHEREXWHOK	NOWSTHEEXACTE
FLGGTEZ?FKZBSFDQVGO	GIPUFXHHRKF	OCATION?ONLYWWTHISW	ASHISLASTMES
FHQNTGUAECNUVPDJMQ	CLQUMUNEDFQ	SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE	VENMINUTESESSIXPOINTF	IVESECONDSNO
DNQFMPNZGLFLPMRJQYA	LMGNUMVPDXVKP	RTHSEVENTYSEVENDEGR	EESEIGHTMINU
DQUMEBEDMHDAFMJGZNU	PLGEWJLLAETGO	TESFORTYFOURSECONDS	WESTXLAYER TWO
ENDYAHROHNLSRHEOCTP	EOIBIDYSHNAIA	SLOWLYDESPARATLYSLO	WESTHLAYERMIN
CHTNREYULDSL LSLNOH	SNOSMRWXMNE	OFFPASSAGEDEBRISTHAT	ENCUMBEREDT
TPRNGATIHNRRAPESLNN	ELEBLPIIACAE	HELOWERPARTOFTHEDOO	RWAYWASREMOV
WMTWNDITEENRAHCTENE	UDRETNHAEOE	EDWITH TREMBLINGHAND	SIMADEATINY
TFOLSEDTIWENHAEIOYT	EYQHEENCTAYCR	BREACHINTHEUPPERLEF	THANDCORNERAN
EIFTBRSPAMHHEWENATA	MATEGYEERLB	DTHENWIDENINGTHEHOL	EALITTLEIIN
TEEFOASF IOTUETUAEO	OARMAEERTNRTI	SERTEDTHECANDLEANDP	EEREDINTHEHOT
BSEDDNIAAHTTMSTEWPI	EROAGRIEWFE	AIRESCAPINGFROMTHEC	HAMBERCAUSED
AECTDDHILCEIHSITEGO	EAOSDDRYDLORIT	THEFLAMETOFLICKERBU	TPRESENTLYDETA
RKLMLEHAGTDHARDPNEO	HMGFMFEUHE	ILSOFTHEROOMWITHINE	MERGEDFROM
ECDMRI PFEIMEHNLSTT	RTVDOHW?OBKR	THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
UOXOGHULBSOLIFBBWFL	RVQQPRNGKSSO	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXX
TWTQSJQSSEKZZWATJKL	UDIAWINFBNYP	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXX
VTTMZFPKWGDZXTJCDI	GKUHUAEKCAR	LINXXXXXXXXXXXXXXXX	XXXXXXXXXXXX

As they say in all the best adventures, 'X marks the spot'. It will be noted that even with the row separation the centre of the X lies between WW. The coordinates were his last message, implying that there is something significant that ties his last position to the properties of the coordinate blocks. This leads us directly to the existence of a concealed row addressing scheme.

Row Address Scheme

The row address scheme discovered is centred round the central block of 77 characters, but does in fact define 14 rows each containing 12 characters. Each row of seven characters in the block of 77 corresponds to 7 characters in the string of K4. If we assume that this addressing scheme starts from the previously found X marker for WW's last position then it is hypothesised that the Kryptos row scheme extends upwards in boustrophedon format, whereby the rows reverse the direction that they should be read off in at the end of each row. This principle will be demonstrated graphically here.



(Rows participating in palimpsest scheme read off in boustrophedon order)

The overrun from block 77 into blocks 57 and 38 becomes more interesting when one examines the underlying mathematical relationship.

$$77 - 57 = 20$$

$$77 + 20 = 97$$

Whether the final 20 rows have been addressed correctly will no doubt be seen in time, but for now it is assumed that the rows continue unbroken in the vertical plane.

Part III

Potential Masking and Substitution Schemes

Now that the row addressing scheme has been outlined, it will be shown how a number of clues within the plain texts point to a potential masking and substitution scheme. It will then be possible to show how such schemes can be used to derive the plain text BERLIN from positions 64 through to 69.

The first obvious move is to number the addresses for each of the rows.

SHADINGANDTH	92	-	98	BETWEENSUBTLE
EOFLIGHTLIES	91	-	85	CEOFIQLUSION
ITWASTOTALLY	78	-	84	LEHOWSTHATPO
SSIBLE?THEYU	77	-	71	ARTHSMAGNET
ICFIELDXTHEI	64	-	70	IONWASGATHER
EDANDTRANSMI	63	-	57	ERGRUUNDTOANU
NKNOWNLOCATI	50	-	56	LANGLEYKNOWA
BOUTTHIS?THE	49	-	43	ITSBURIEDOUT
THERESOMEWHE	36	-	42	NOWSTHEEXACTL
QCATION?ONLY	35	-	29	ASHISLASTMES
SAGEXTHIRTYE	22	-	28	REESFIFTYSE
VENMINUTESSE	21	-	15	IVESECONDSNO
RTHSEVENTYSE	08	-	14	ESEIGHTMINU
TESFORTYFOUR	07	-	01	WESTXLAYERTWO

The obvious row of interest is 64 – 70 as we are looking to find the decryption in plaintext of BERLIN in positions 64 – 69. Just as with the null cipher earlier we find that an encryption with the Vigenère scheme using NYPVTT yields ICXUEI.

If we compare the row of plaintext from K2 on the right hand side in this position then we find ICFIELDXTHEI. Masking the word FIELD and substituting the TH for a U would give us our necessary target, but we need to find some supporting evidence that such a scheme is in place as well as determine the nature of its behaviour.

Possible Masking Scheme for rows proceeding left to right

The word FIELD was originally identified as a potential mask by using an algorithm that looked for Vigenère encryption or decryption relations between keywords in the plaintext or cipher text. The relation found here was originally alluded to by the linkage of an anagram of QUXQB (This is the word LIGHT in the cipher text) to FIELD and AGNET in the plain text. Once attention was centred in this area the next clue became apparent.

Looking at the previously discovered text strings for HOT in the vertical plane it was seen that these could be extended to give obfuscated anagrams of NORTH and SOUTH whereby the U of South had been replaced by a C and the R of North replaced by a T.

USDJKZLDKRNSHGNFIVJ	EMUFPHZLRFAXY	SHADINGANDTHEABSENC	BETWEENSUBTLE
YQTQUXQBQVYUULLTREV	JYQTMKYRDMFD	EOFLIGHTLIESTHENUAN	CHOFIQLUSION
VFPJUDEEHZWETZYVGWH	KKQETGFQJNCE	ITWASTOTALLYINVISIB	LEHOWSTHATPO
GGWHKK?DQMCPPQZDQMM	IAGPFXHQRLG	SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA	ICFIELDXTHEINFORMAT	IONWASGATHER
QZGZLECGYUXUEENJTBJ	LBQCRTBJDFHRR	EDANDTRANSMITTEDUND	ERGRUUNDTOANU
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI	NKNOWNLOCATIONXDOES	LANGLEYKNOWA
HHDDDUVH?DWKBFUFPPWN	TDFIYCUQZERE	BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
EVLDKFEZMOQQJLTTUGS	YQPFUNLAVIDX	THERESOMEWHEREXWHOK	NOWSTHEEXACTL
FLGGTEZ?FKZBSFDQVGO	GIPUFXXHHRKF	OCATION?ONLYWWTHISW	ASHISLASTMES
FHQNTGPUAECNUVPDJMQ	CLQUMUNEDFQ	SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE	VENMINUTESSIXPOINTF	IVESECONDSNO
DNQFMPNZGLFLPMRJQYA	LMGNUMPDXXVKP	RTHSEVENTYSEVENDEGR	ESEIGHTMINU
DQUMEBEDMHDAFMJGZNU	PLGEWJLLAETGO	TESFORTYFOURSECONDS	WESTXLAYERTWO
ENDYAHROHNLSRHEOCP	EOIBIDYSHNAIA	SLOWLYDESPARATLYSLO	WESTHLAYERMIN
CHTNREYULDSL LSL LNOH	SNOSMRWXMNE	OPPASSAGEDEBRISTHAT	ENCUMBEREDT
TPRNGATIHNRRARPESLNN	ELEBLPIIACAE	HELOWERPARTOFTHEDOO	RWAYWASREMOV
WMTWNDITEENRAHCTENE	UDRETNHAEOE	EDWITH TREMBLINGHAND	SIMADEATINY
TFOLSEDTIWENHAEIOYT	EYQHEENCTAYCR	BREACHINTHEUPPERLEF	THANDCORNERAN
EIFTBRSPAMHHEWENATA	MATEGYEERLB	DTHENWIDENINGTHEHOL	EALITTLEIIN
TEEFOASF IOTUETUAEOT	OARMAEERTNRTI	SERTEDTHECANDLEANDP	EEREDINTHEHOT
BSEDDNIAAHTTMSTEWPI	EROAGRIEWFEB	AIRESCAPINGFROMTHEC	HAMBERCAUSED
AECTDDHILCEIHSITEGO	EAOSDDRYDLORIT	THEFLAMETOFLICKERBU	TPRESENTLYDETA
RKLMLHAGTDHARDPNEO	HMGFMFEUHE	ILSOFTHEROOMWITHINE	MERGEDFROM
ECDMRI PFEIMEHNL SSTT	RTVDOHW?OBKR	THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
UOXOGHULBSOLIFBBWFL	RVQQPRNGKSSO	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXXX
TWTQSJQSSEKZZWATJKL	UDIAWINFBNYP	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXBER
VTTMZFPKWGDKZXTJCDI	GKUHUAAUEKCAR	LINXXXXXXXXXXXXXXXX	XXXXXXXXXXXXXX

It is hypothesised that this is the structure alluded to by the 'EARTHS MAGNETIC FIELD' clue in K2. If we do another steganographic transform to move the right hand block to the left hand then this becomes even more of stark clue. It will also be seen that the word WEST in the bottom right now also resides in the WEST half of the table.

The string 'HOT' represents the letters in common to both NORTH and SOUTH. It will be remembered from child hood that a MAGNET will have the effect of repelling or attracting a NORTH and SOUTH pole in the presence of its field. Full stops have been used to show the relationship with the gaps that appear when the text is repositioned in such a manner. The appendices at the back of this document will discuss the potential pattern of dots and there relation to Morse encodings.

B	E	T	W	E	E	N	S	U	B	T	L	E	S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	
C	E	O	F	I	Q	L	U	S	I	O	N	.	E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	
L	E	H	O	W	S	T	H	A	T	P	O	.	I	T	W	A	S	T	O	T	A	L	L	I	N	V	I	S	I	B		
A	R	T	H	S	M	A	G	N	E	T	.	.	S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	
I	C	N	W	A	S	G	A	T	H	E	R	.	I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	
E	R	G	R	U	N	D	T	O	A	N	U	E	D	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D		
L	A	N	G	L	E	Y	K	N	O	W	A	.	N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	
I	T	S	B	U	R	I	E	D	O	U	T	.	B	O	U	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D		
N	O	W	S	T	H	E	E	X	A	C	T	L	T	H	E	R	E	S	S	O	M	E	W	H	E	R	E	X	W	H	O	K
A	S	H	I	S	L	A	S	T	M	E	S	.	O	C	A	T	I	O	N	?	O	N	L	Y	W	W	T	H	I	S	W	
R	E	E	S	F	I	F	T	Y	S	E	.	.	S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	
I	V	E	S	E	C	O	N	D	S	N	O	.	V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	
E	E	S	E	I	G	H	T	M	I	N	U	.	R	T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	
W	E	S	T	X	L	A	Y	E	R	T	W	O	T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	

The fact that each of the poles is missing a letter suddenly made it clear that in the non palimpsest text area in the row 64-70 there were enough letters to mark SO(U)TH, but without a U.

I (O) N W A (S) G A (T) (H) E R

It turned out that all the other cardinal points of the compass could be constructed from this string, so a trawl was made of the relationships between the redundant block on the left and all the possible cardinal points that could be extracted.

The resulting set of relationships for each row is as follows.

SHADINGANDTHEABSENC	BETWEENSUBTLE	NORTH	EAST	SOUTH	WEST
EOFLIGHTLIESTHENUAN	CEOFIQLUSION	NORTH	EAST	SOUTH	WEST
ITWASTOTALLYINVISIB	LEHOWSTHATPO	NORTH	EAST	SOUTH	WEST
SSIBLE?THEYUSEDTHEE	ARTHSMAGNET	NORTH	EAST	SOUTH	WEST
ICFIELDXTHEINFORMAT	IONWASGATHER	NORTH	EAST	SOUTH	WEST
EDANDTRANSMITTEDUND	ERGRUUNDTOANU	NORTH	EAST	SOUTH	WEST
NKNOWNLOCATIONXDOES	LANGLEYKNOWA	NORTH	EAST	SOUTH	WEST
BOUTTHIS?THEYSHOULD	ITSBURIEDOUT	NORTH	EAST	SOUTH	WEST
THERESOMEWHEREXWHOK	NOWSTHEEXACTL	NORTH	EAST	SOUTH	WEST
OCATION?ONLYWWTHISW	ASHISLASTMES	NORTH	EAST	SOUTH	WEST
SAGEXTHIRTYEIGHTDEG	REESFIFTYSE	NORTH	EAST	SOUTH	WEST
VENMINUTESEXPOINTF	IVESECONDSNO	NORTH	EAST	SOUTH	WEST
RTHSEVENTYSEVENDEGR	ESEIGHTMINU	NORTH	EAST	SOUTH	WEST
TESFORTYFOURSECONDS	WESTXLAYERTWO	NORTH	EAST	SOUTH	WEST
SLOWLYDESPARATLYSLO	WESTHLAYERMIN	NORTH	EAST	SOUTH	WEST
OPPASSAGEDEBRISTHAT	ENCUMBEREDT	NORTH	EAST	SOUTH	WEST
HELOWERPARTOFTHEDOO	RWAYWASREMOV	NORTH	EAST	SOUTH	WEST
EDWITHTREMBLINGHAND	SIMADEATINY	NORTH	EAST	SOUTH	WEST
BREACHINTHEUPPERLEF	THANDCORNERAN	NORTH	EAST	SOUTH	WEST
DTHENWIDENINGTHEHOL	EALITTLEIIN	NORTH	EAST	SOUTH	WEST
SERTEDTHECANDLEANDP	EEEREDINTHEHOT	NORTH	EAST	SOUTH	WEST
AIRESCAPINGFROMTHEC	HAMBERCAUSED	NORTH	EAST	SOUTH	WEST
THEFLAMETOFLICKERBU	TPRESENTLYDETA	NORTH	EAST	SOUTH	WEST
ILSOFTHEROOMWITHINE	MERGEDFROM	NORTH	EAST	SOUTH	WEST
THEMISTXCANYOUSEEAN	YTHINGQ?	NORTH	EAST	SOUTH	WEST

The most striking aspect of this analysis is that only this one row exhibits this relationship, whereby all cardinal points of the compass are reproducible from the (redundant?) plain text.

The reward of any intellectual endeavour is always that *eureka moment* when all the different strands that have been kept up in the air finally come together to form a coherent whole that is instantly recognisable as being beyond chance. Whilst running the afore mentioned algorithm to explore relationships between discrete sections of the plain text and cipher text I found a region that did not code at all for the strings LIGHT or QUXQB whereas every other position in the string did (I will expand on this algorithm in the appendices). It is thought that this may bear some significance relating to the clue, THE ABSENCE OF LIGHT.

I should also point out that the line I was parsing was the equivalent of rows 64 – 70, but in the reflected K3 rows that are in the half below the X-axis.

Amazingly I didn't see the most striking relationship at first glance, so I will ask the reader to see if they spot it on the next page before making the relationship explicit. Remember that this string that didn't code for LIGHT or QUXQB was found by computational analysis and not by eye.

Result of computational parsing for strings that do not relate through anagrammatic translation in any way to either QUXQB or LIGHT when fed the symmetric string from the corresponding row above the X-axis. This is the result for rows 64-70 in K2 and the symmetric equivalent in K3.

SHADINGANDTHEABSENC	BETWEENSUBTLE
EOFLIGHTLIESTHENUAN	CEOFIQLUSION
ITWASTOTALLYINVISIB	LEHOWSTHATPO
SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
IOFIELDXTHEINFORMAT	IONWASGATHER
EDANDTRANSMITTEDUND	ERGRUUNDTOANU
NKNOWNLOCATIONXDOES	LANGLEYKNOWA
BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
THERESOMEWHEREXWHOK	NOWSTHEEXACTL
OCATION?ONLYWWTHISW	ASHISLASTMES
SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
VENMINUTESEXPOINTF	IVESECONDSNO
RTHSEVENTYSEVENDEGR	ESEIGHTMINU
TESFORTYFOURSECONDS	WESTXLAYERTWO
SLOWLYDESPARATLYSLO	WESTHLAYERMIN
OFPASSAGEDEBRISTHAT	ENCUMBEREDT
HELOWERPARTOFTHEDOO	RWAYWASREMOV
EDWITHTREMBLINGHAND	SIMADEATINY
BREACHINTHEUPPERLEF	THANDCORNERAN
DTHENWIDENINGTHEHOL	EALITTLEIIN
SERTEDTHECANDLEANDP	EEREDINTHEHOT
AIRESCAPINGFROMTHEC	HAMBERCAUSED
THEFLAMETOFLICKERBU	TPRESENTLYDETA
II(SOFTHEROOMWITHINE	MERGEDFROM
THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
XXXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXBER
LINXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXX

I shall let the reader reflect on this page and continue on the next.

To make explicit what was found on the last page look at the letters highlighted in the K3 row. They read SOFTH. In other words (or letters even) SO(U)TH. On re-examining what was deemed to be the redundant information in row 64-70 we find the following string SAGTH. Putting this all together we end up with what looks like a triangular relationship between the masked string and the word that has been used as a substitution target.

SHADINGANDTHEABSENC	BETWEENSUBTLE
EOFLIGHTLIESTHENUAN	CEOFIQLUSION
ITWASTOTALLYINVISIB	LEHOWSTHATPO
SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
ICFIELDKTHEINFORMAT	IONASGATHER
EDANTTRANSMITTEDUND	ERGRUUNTOANU
NKNOWNLOCATIONXDOES	LANGLEYKNOWA
BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
THERESOMEWHEREXWHOK	NOWSTHEEXACTL
OCATION?ONLYWWTHISW	ASHISLASTMES
SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
VENMINUTESEXPOINTF	IVESECONDSNO
RTHSEVENTYSEVENDEGR	EESEIGHTMINU
TESFORTYFOURSECONDS	WESTXLAYERTWO
SLOWLYDESPARATLYSLO	WESTHLAYERMIN
OFPASSAGEDEBRISTHAT	ENCUMBEREDT
HELOWERPARTOFTHEDOO	RWAYWASREMOV
EDWITHTREMBLINGHAND	SIMADEATINY
BREACHINTHEUPPERLEF	THANDCORNERAN
DTHENWIDEMINGTHEHOL	EALITTLEIIN
SERTEDTHECANDLEANDP	EEREDINTHEHOT
AIRESCAPINGFROMTHEC	HAMBERCAUSED
THEFLAMETOFLICKERBU	TPRESENTLYDETA
II[SOFTH]EROOMWITHINE	MERGE[DFROM]
THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
XXXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXBER
LINXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXX

If we now allow ourselves to swap letters within the confines of a single row then we can arrive at the following arrangement for row 64- 70 and its symmetric equivalent in K3.

SHADINGANDTHEABSENC	BETWEENSUBTLE
EOFLIGHTLIESTHENUAN	CEOFIQLUSION
ITWASTOTALLYINVISIB	LEHOWSTHATPO
SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
ICFIELDXTHEINAGRMAT	IONASOFTHER
EDANDTRANSMITTEDUND	ERGRUUNTOANU
NKNOWNLOCATIONXDOES	LANGLEYKNOWA
BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
THERESOMEWHEREXWHOK	NOWSTHEEXACTL
OCATION?ONLYWWTHISW	ASHISLASTMES
SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
VENMINUTESSEXPOINTF	IVESECONDSNO
RTHSEVENTYSEVENDEGR	EESEIGHTMINU
TESFORTYFOURSECONDS	WESTXLAYERTWO
SLOWLYDESPARATLYSLO	WESTHLAYERMIN
OPASSAGEDEBRISTHAT	ENCUMBEREDT
HELOWERPARTOFTHEDOO	RWAYWASREMOV
EDWITHTREMBLINGHAND	SIMADEATINY
BREACHINTHEUPPERLEF	THANDCORNERAN
DTHENWIDENINGTHEHOL	EALITTLEIIN
SERTEDTHECANDLEANDP	EEREDINTHEHOT
AIRESCAPINGFROMTHEC	HAMBERCAUSED
THEFLAMETOFLICKERBU	TPRESENTLYDETA
RO(SOFTH)NOOMWITHINE	MERCE(FIELD)
THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
XXXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXBER
LINXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXXX

A bet is now hedged with regards the substitution scheme whereby the TH is transformed to a U. It may be that the scheme is as simple as looking for the presence of any parts of the masking keyword in the palimpsest section of the plain text and then replacing with the missing letters required to make the correct word in this case the U of SO(U)TH. The letter replaced would then be appended to the end of the line to make up our full block of 7. Our final string in this instance would be ICXUEIF which in turn would give a plain text of BERLINT.

An alternate scheme is now given whereby the letters that are substituted for the missing letters of the masking keyword are deemed significant. Remember that FO was swapped for GA in order to arrive at SOFTH on the right hand side. If we put HT through the Vigenère decryption process along with AG then we arrive at UL. If U and L are the letters we are substituting then making up the row to 7 now gives ICXUEIL. This particular string is of interest as we have already inadvertently stumbled across the string UEIL already. It was mentioned on page 21 that the gaps in the row endings may have some significance in Morse Code. If we examine the relation found which is: .._ . . . _.. then we can translate to plain text to find the string UEIL. This string would give a plain text of BERLINC.

Possible Masking Scheme for rows proceeding right to left

Once the start point and text direction was established, a credible method for decrypting rows right to left was established that followed the exact same pattern and method as outlined above. The difference with the right to left scheme is that the masking appears to be different, which may mean that each individual row has its own discrete method of masking and substitution. Rather than wait to find all the discrete masks and substitutions for each row, I thought it better to publish in the hope that any errors in this work can be corrected by the wider community and that any clues missed can be brought to bear on the same underlying palimpsest and row schematic in order to arrive more quickly at a complete decryption. Please feel free to take this paper as a whole or extract various clues in order to develop your own decryption schemes, but please cite this original paper in any subsequent write ups.

With regards the row representing string positions 1 – 7, the following observation was made.

SHADINGANDTHEABSENC
EOFLIGHTLIESTHENUAN
ITWASTOTALLYINVISIB
SSIBLE?THEYUSEDTHEE
ICFIELDXTHEINFORMAT
EDANDTRANSMITTEDUND
NKNOWNLOCATIONXDOES
BOUTTHIS?THEYSHOULD
THERESOMEWHEREEXWHOK
OCATION?ONLYWWTHISW
SAGEXTHIRTYEIGHTDEG
VENMINUTESEXPOINTF
RTHSEVENTYSEVENDEGR
TESFORTYFOURSECONDS

There is the potential to apply a mask in this row that looks incredibly similar to one of our first keys which was CE OF. Remembering to use SUBTLE SHADING to occlude, we can arrive at the interchangeable strings of OF and OF. Remember also that the previous decryption scheme of right to left also involved the use of the string FO being swapped to give OF.

To find some reasoning that is vaguely consistent with the proposed right to left scheme outlined above it is possible to look slightly further and find a possible method that gives an additional explanation for the use of this mask.

USDJKZLDKRNSHGNFIVJ	EMUFPHZLRFAXY	SHADINGANDTHEABSENC	BETWEENSUBTLE
YQTQUXQBQVYUULLTREV	JYQTMKYRDMFD	EOFLIGHTLIESTHENUAN	CEOFIQLUSION
VFPJUDEEHZWETZYVGWH	KKQETGFQJNCE	ITWASTOTALLYINVISIB	LEHOWSTHATPO
GGWHKK?DQMCPFQZDQMM	IAGPFXHQRLG	SSIBLE?THEYUSEDTHEE	ARTHSMAGNET
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA	ICFIELDXTHEINFORMAT	IONWASGATHER
QZGZLECGYUXUEENJTBJ	LBQCRTBJDFHRR	EDANDTRANSMITTEDUND	ERGRUUNDTOANU
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI	NKNOWNLOCATIONXDOES	LANGLEYKNOWA
HHDDDUVH?DWKBFUFPNW	TDFIYCUQZERE	BOUTTHIS?THEYSHOULD	ITSBURIEDOUT
EVLDKFEZMOQQJLTUGS	YQPFEUNLAVIDX	THERESOMEWHEREXWHOK	NOWSTHEEXACTL
FLGGTEZ?FKZBSFDQVGO	GIPUFXXHHRKF	OCATION?ONLYWWTHISW	ASHISLASTMES
FHQNTGPAECNUVPDJMQ	CLQUMUNEDFQ	SAGEXTHIRTYEIGHTDEG	REESFIFTYSE
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE	VENMINUTESSEXPOINTF	IVESECONDSNO
DNQFMPNZGLFLPMRJQYA	LMGNUMVPDXVKP	RTHSEVENTYSEVENDEGR	EESIEIGHTMINU
DQUMEBEDMHDAFMJGZNU	PLGEWJLLAETGO	TESFORTYFOURSECONDS	WESTXLAYERTWO
ENIYAHROHNLSRHEOCP	EOIBIDYSHNAIA	SLOWLYDESPARATLYSLO	WESTHLAYERMIN
CHTNREYULDSL LSLNOH	SNOSMRWXMNE	OPPASSAGEDEBRISTHAT	ENCUMBEREDT
TPRNGATIHNRRAPESLNN	ELEBLPIIACAE	HELOWERPARTOFTHEDOO	RWAYWASREMOV
WMTWNDITEENRAHCTENE	UDRETNHAEOE	EDWITHTREMBLINGHAND	SIMADEATINY
TFOLSEDTIWENHAEIOYT	EYQHEENCTAYCR	BREACHINTHEUPPERLEF	THANDCORNERAN
EIFTBRSPAMHHEWENATA	MATEGYEERLB	DTHENWIDENINGTHEHOL	EALITTLEIIN
TEEFOASF IOTUETUAEOT	OARMAERTNRTI	SERTEDTHECANDLEANDP	EEREDINTHEHOT
BSEDDNIAAHTTMSTEWPI	EROAGRIEWFEB	AIRESCAPINGFROMTHEC	HAMBERCAUSED
AECTDDHILCEIHSITEGO	EAOSDDRYDLORIT	THEFLAMETOFLICKERBU	TPRESENTLYDETA
RKLMLEHAGTDHARDPNEO	HMGFMFEUHE	ILSOFTHEROOMWITHINE	MERGEDFROM
ECDMRI PF EIMEHNLSSTT	RTVDOHW?OBKR	THEMISTXCANYOUSEEAN	YTHINGQ?XXXX
UOXOGHULBSOLIFBBWFL	RVQQPRNGKSSO	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXX
TWTQSJSSEKZZWATJKL	UDIAWINFBNYP	XXXXXXXXXXXXXXXXXXXX	XXXXXXXXXXXX
VTTMZFPKWGDKZXTJCDI	GKUHUAEUKCAR	LINXXXXXXXXXXXXXXXX	XXXXXXXXXXXX

Using the Vigenère method we can decrypt OFOF by YAHHR to derive the word POLE. This is slightly more tenuous than the method used to derive FIELD, but even if a coincidence it was felt too noteworthy to leave out of this final draft.

Now that we believe we have applied the right mask, we need to find a suitable substitution. In this instance *the substitution was guessed* to be RT -> P.

Substituting ET for P and reading from right to left gives the string RUYTRSP. When this string is encrypted with the first 7 letters of the K4 plain text, which are OBKRUX we derive a string of:

SPY OVER

Conclusion

It has been shown that there is a potential palimpsest in K1 and K2 that with appropriate masking and substitution may yield the final K4 plain text when encrypted with the known K4 cipher text. For every clue found in the preceding sections there are no doubt many more that remain hidden, but the existence of a coherent framework underlying the decryption process should dramatically reduce the time and scope of the algorithms needed in order to flush these out and find a complete decryption. There are bound to be many mistakes and wrong turns made in this paper, but I'm hopeful that the row addressing scheme in itself is a meaningful contribution that will feature in any complete decryption.

This paper is no doubt likely to cause some lively debate, but I would urge community members to keep this within the bounds of the Kryptos group forum. Should anyone want to discuss any of the ideas in this paper in more detail then please join the forum and find me under my user name of matzreid888@yahoo.co.uk.

Appendices

Appendix A.

Elemental Substitution scheme

It was noted that the substitution scheme may have something to do with elements and their underlying atomic number. This was based on three observations, the first being that the substitution relationship Th->U may be a nod towards the Thorium to Uranium transmutation process.

Te to P is more tenuous, but we do have a very interesting artefact in the cipher text that would show a method for collapsing two letters in to one. In this example for gold (Au) it is conjectured that Pt would be substituted for A. The reasoning behind this is highlighted below.

USDJKZLDKRNSHGNFIVJ	EMUFPHZLRFAXY
YQTQUXQBQVYUULLTREV	JYQTMKYRDMFD
VFPJUDEEHZWETZYVGWH	KKQETGFQJNCE
GGWHKK?DQMCPFQZDQMM	IAGPFXHQRLG
TIMVMZJANQLVKQEDAGD	VFRPJUNGEUNA
QZGZLECGYUXUEENJTBJ	LBQCRTBJDFHRR
YIZETKZEMVDUFKSJHKF	WHKUWQLSZFTI
HHDDDUVH?DWKBFUFPNW	TDFIYCUQZERE
EVLDKFEZMOQQJLTTUGS	YQPFEUNLAVIDX
FLGGTEZ?FKZBSFDQVGO	GIPUFXXHHRKF
FHQNTGPAECNUVPDJMQ	CLQUMUNEDFQ
ELZZVRRGKFFVOEEXBDM	VPNFQXEZLGRE
DNQFMPNZGLFLPMRJQYA	LMGNUMPD XVKP
DQUMEBEDMHDAFMJGZNU	PLGEWJLLAETGO
ENDYAHROHNLSRHEOCPT	EOIBIDYSHNAIA
CHTNREYULDSLSLLNOH	SNOSMRWXMNE

Pt and Au are in positions 78 and 79 of the periodic table, which is why this was deemed significant. It may be that there is still a substitution table still to be discovered or that there is a direct link to the periodic system.

Appendix B.

Embedded Morse?

In the swap of the right hand block to the left hand side it was seen that the spaces seemed to follow a pattern. Whether or not this is meaningful Morse Code or some other code remains to be seen, but the following observation was made.

For the upper half (K1 & K2) it can be seen that the spaces code to the symmetrical string:

••__ •/••/•__••

and that the bottom half (K3) codes to:

•• • ••/••/•(•••) •

(The 3 dots in parenthesis are left as 3 single dots rather than a guess made as to the permutation of dot to dash).

Appendix C.

Algorithms used

The algorithms used throughout were purely the standard encryption and decryption methods that utilised the Vigenère tableau built into the sculpture.

New algorithms were only needed in order to ease the finding of strings in the vertical plane of the texts and looking for any anagrammatic relationships that may decode from one part of the text into another.